Linear Algebra and Its Applications

Section 1.6: Applications of Linear Systems

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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1.6 Applications of linear systems

Three modeling ideas organize this section.

- In an exchange economy, column j of the exchange table records where sector j's output goes. Each column sums to 1. If E is this table and p lists the values of total outputs, equilibrium means $Ep = p$, or $(E - I)p = 0$.
- In a chemical reaction, conserve each type of atom separately. Assign positive unknown coefficients to all compounds; write reactant atom counts minus product atom counts equal to zero. Choose the smallest positive integer solution.
- In a flow network, write incoming flow equals outgoing flow at every node. After solving the linear equations, apply the required nonnegativity inequalities. A free flow parameter is not automatically allowed to take negative values.

Practice problem 1.6.P1: Build the exchange table in the stated order

Use the order Agriculture, Mining, Manufacturing for both rows and columns. The three columns are respectively $(.65, .05, .30)^T$, $(.20, .10, .70)^T$, and $(.20, .30, .50)^T$:

$$E = \begin{bmatrix} .65 & .20 & .20 \\ .05 & .10 & .30 \\ .30 & .70 & .50 \end{bmatrix}.$$

For example Agriculture retains $1 - .05 - .30 = .65$, so that value belongs in the first entry of the first column. Each displayed column sums to 1; this checks that all output was allocated exactly once.

Practice problem 1.6.P2: Use flow inequalities after solving the equations

Example 2 gives $x_1 = 600 - x_5$, $x_2 = 200 + x_5$, $x_3 = 400$, $x_4 = 500 - x_5$. Nonnegative x_5 and x_4 imply $0 \le x_5 \le 500$. The remaining two inequalities are then automatic. Consequently $100 \le x_1 \le 600$ and $200 \le x_2 \le 700$. Both endpoints occur, at $x_5 = 500$ and $x_5 = 0$, respectively. The two flows vary together: as x_1 decreases, x_2 increases.

Exercise 1.6.1: Two-sector equilibrium is a ratio, not a unique price list

Use the order Goods, Services. The exchange matrix and equations are

$$E = \begin{bmatrix} .20 & .70 \\ .80 & .30 \end{bmatrix}, \qquad .20p_G + .70p_S = p_G, \quad .80p_G + .30p_S = p_S.$$

Both equations simplify to $8p_G = 7p_S$. Thus

$$p = t(7, 8)^T, \qquad t > 0.$$

For example output values 7 and 8 work: Goods spends $.20(7) + .70(8) = 7$, while Services spends $.80(7) + .30(8) = 8$.

Conclusion. Any common positive multiple of $(7, 8)^T$ is an equilibrium.

Exercise 1.6.2: A currency change rescales every output value

In the Coal–Electric–Steel order of Example 1, the exact equations are

$$p_C = .4p_E + .6p_S, \qquad .9p_E = .6p_C + .2p_S.$$

Insert the first into the second: $.9p_E = .24p_E + .56p_S$, so $.66p_E = .56p_S$ and $p_E = (28/33)p_S$. Then $p_C = (31/33)p_S$. Thus an exact alternative is $(p_C, p_E, p_S) = (62, 56, 66)$. Checking the third row gives $.4(62) + .5(56) + .2(66) = 66$. The example rounds its decimal answer; these rational ratios avoid that rounding error. If one dollar is $k > 0$ yen, every component is multiplied by k. Because $E(kp) = kEp = kp$, the exchange proportions and equilibrium relations do not change, only the numerical currency units do.

Exercise 1.6.3: A three-sector exchange economy

(a) Place each sector's sales in its column.

In the order Fuels and Power, Manufacturing, Services,

$$E = \begin{bmatrix} .10 & .10 & .20 \\ .80 & .10 & .40 \\ .10 & .80 & .40 \end{bmatrix}.$$

The retained amounts are $.10, .10, .40$, respectively. Each column sums to 1.

(b) Set the output value equal to the cost of purchased inputs.

Writing the values as f, m, s, the equations $Ep = p$ are

$$.10f + .10m + .20s = f, \quad .80f + .10m + .40s = m, \quad .10f + .80m + .40s = s.$$

Multiplying $(I - E)p = 0$ by 10 gives the augmented matrix

$$\left[\begin{array}{rrr|c} 9 & -1 & -2 & 0 \\ -8 & 9 & -4 & 0 \\ -1 & -8 & 6 & 0 \end{array}\right].$$

(c) Set the Services value to 100 and solve exactly.

The first equation gives $m = 9f - 2s$. Insert this in the second: $-8f + 9(9f - 2s) - 4s = 0$, hence $73f = 22s$. Thus $f = 22s/73$ and $m = 52s/73$. For $s = 100$,

$$p = (2200/73, 5200/73, 100)^T \approx (30.136986, 71.232877, 100)^T.$$

The third equation checks because $-22 - 8(52) + 6(73) = 0$ after factoring out $s/73$.

Exercise 1.6.4: An exact four-sector equilibrium

(a) Construct the exchange matrix.

Use Mining, Lumber, Energy, Transportation in that order:

$$E = \begin{bmatrix} .30 & .15 & .20 & .20 \\ .10 & .15 & .15 & .10 \\ .60 & .50 & .45 & .50 \\ 0 & .20 & .20 & .20 \end{bmatrix}.$$

For instance Mining retains $1-.10-.60=.30$, and the Energy column retains $1-.20-.15-.20=.45$.

(b) Solve $(I-E)p=0$.

Multiplying by 20 avoids decimals. The coefficient matrix is

$$20(I-E) = \begin{bmatrix} 14 & -3 & -4 & -4 \\ -2 & 17 & -3 & -2 \\ -12 & -10 & 11 & -10 \\ 0 & -4 & -4 & 16 \end{bmatrix}.$$

Its reduced equations are

$$p_M = \frac{193}{141}p_T, \qquad p_L = \frac{118}{141}p_T, \qquad p_E = \frac{446}{141}p_T.$$

For clarity one can obtain these ratios without decimal arithmetic: the last equation gives $p_L + p_E = 4p_T$. Substituting $p_E = 4p_T - p_L$ in the second gives $20p_L = 2p_M + 14p_T$; the first gives $14p_M + p_L = 20p_T$. Solving these two equations gives the displayed ratios. Choose $p_T = 141$. Then $p = (193, 118, 446, 141)^T$ is positive. The rows of $20(I-E)p$ are $2702 - 354 - 1784 - 564 = 0$, $-386 + 2006 - 1338 - 282 = 0$, $-2316 - 1180 + 4906 - 1410 = 0$, and $-472 - 1784 + 2256 = 0$.

Conclusion. One exact equilibrium is $(193, 118, 446, 141)^T$; every positive common multiple also works.

Exercise 1.6.5: Compare the equilibrium before and after a change in trade

(a) Build the original table.

Use Agriculture, Manufacturing, Services, Transportation:

$$E = \begin{bmatrix} .20 & .35 & .10 & .20 \\ .20 & .10 & .20 & .30 \\ .30 & .35 & .50 & .20 \\ .30 & .20 & .20 & .30 \end{bmatrix}.$$

Each column is the allocation of the sector at its top, including its retained share on the diagonal.

(b) Normalize Transportation to 10.

Writing a, m, s, t for the four output values, multiply $(I-E)p=0$ by 20. The equations are

$$\begin{aligned} 16a - 7m - 2s - 4t &= 0, & -4a + 18m - 4s - 6t &= 0, \\ -6a - 7m + 10s - 4t &= 0, & -6a - 4m - 4s + 14t &= 0. \end{aligned}$$

The second and fourth give $2m + 2s = 7t - 3a$ and $-2a + 9m - 2s = 3t$, so $11m = 10t - a$. The first equation together with $2m + 2s = 7t - 3a$ gives $19a - 5m = 11t$. Substitute $11m = 10t - a$ to

obtain $209a - 5(10t - a) = 121t$, or $214a = 171t$. Therefore $a = (171/214)t$, $m = (179/214)t$, and $s = (627/428)t$. For $t = 10$,

$$p = (855/107, 895/107, 3135/214, 10)^T \approx (7.990654, 8.364486, 14.649533, 10)^T.$$

(c) Change just the Agriculture column.

The new shares from Agriculture are $.20, .10, .40, .30$; the other columns stay the same. Thus

$$\widetilde{E} = \begin{bmatrix} .20 & .35 & .10 & .20 \\ .10 & .10 & .20 & .30 \\ .40 & .35 & .50 & .20 \\ .30 & .20 & .20 & .30 \end{bmatrix}.$$

(d) Solve and compare with the same normalization.

The new coefficient matrix is

$$20(I - \widetilde{E}) = \begin{bmatrix} 16 & -7 & -2 & -4 \\ -2 & 18 & -4 & -6 \\ -8 & -7 & 10 & -4 \\ -6 & -4 & -4 & 14 \end{bmatrix}.$$

Its reduced equations give $a = (57/73)t$, $m = (56/73)t$, and $s = (114/73)t$. For example the last row again gives $2m + 2s = 7t - 3a$, and the second becomes $11m = 10t - 2a$; the first still gives $19a - 5m = 11t$. Substitution now yields $209a - 5(10t - 2a) = 121t$, hence $219a = 171t$ and $a = (57/73)t$, from which the other two ratios follow. At $t = 10$ the exact solution is

$$\widetilde{p} = (570/73, 560/73, 1140/73, 10)^T \approx (7.808219, 7.671233, 15.616438, 10)^T.$$

Direct multiplication verifies both $Ep = p$ and $\widetilde{E}\widetilde{p} = \widetilde{p}$. With Transportation fixed, Agriculture and Manufacturing decrease by about 0.182435 and 0.693253, while Services increases by about 0.966905. These are changes in the values of total sector outputs under the model's convention, not independently determined retail prices of individual products.

Exercise 1.6.6: Balance aluminum oxide and carbon

Let the four coefficients, in the printed order, be a, b, c, d. Using atom order Al, O, C gives the vector equation

$$a(2, 3, 0)^T + b(0, 0, 1)^T = c(1, 0, 0)^T + d(0, 2, 1)^T.$$

Thus $2a = c$, $3a = 2d$, and $b = d$. Choose the smallest positive a making all coefficients integral: $a = 2$ gives $d = 3, b = 3, c = 4$.

Conclusion. $2\mathrm{Al_2O_3} + 3\mathrm{C} \longrightarrow 4\mathrm{Al} + 3\mathrm{CO_2}$. The atom totals are Al: $4 = 4$, O: $6 = 6$, C: $3 = 3$.

Exercise 1.6.7: Count every hydrogen in citric acid

Let the coefficients of $\mathrm{NaHCO_3}, \mathrm{H_3C_6H_5O_7}, \mathrm{Na_3C_6H_5O_7}, \mathrm{H_2O}, \mathrm{CO_2}$ be a, b, c, d, e. The citric acid molecule has $3 + 5 = 8$ hydrogen atoms. In atom order Na, H, C, O, the conservation equations are

$$a = 3c, \quad a + 8b = 5c + 2d, \quad a + 6b = 6c + e, \quad 3a + 7b = 7c + d + 2e.$$

This is the coordinate form of the vector equation obtained by listing those four atom counts for each compound. Set $c = 1$, so $a = 3$. The hydrogen equation gives $d = 4b - 1$; the carbon equation gives $e = 6b - 3$. The oxygen equation becomes $9 + 7b = 7 + (4b - 1) + 2(6b - 3) = 16b$, hence $b = 1$. Thus $d = e = 3$.

Conclusion. $3NaHCO_3 + H_3C_6H_5O_7 \longrightarrow Na_3C_6H_5O_7 + 3H_2O + 3CO_2$. Atom totals on both sides are Na: 3, H: 11, C: 9, O: 16.

Exercise 1.6.8: Balance limestone neutralization

Use coefficients a, b, c, d, e in the printed order, and atom order H, O, Ca, C. The vector equation gives

$$3a = 2c, \quad a + 3b = c + 2e, \quad b = d, \quad b = e.$$

Substitute $d = e = b$ in the oxygen equation to get $c = a + b$. Then $3a = 2(a + b)$ gives $a = 2b$. The smallest choice $b = 1$ therefore gives $(a, b, c, d, e) = (2, 1, 3, 1, 1)$.

Conclusion. $2H_3O + CaCO_3 \longrightarrow 3H_2O + Ca + CO_2$. The H, O, Ca, C totals are respectively $6, 5, 1, 1$ on each side.

Exercise 1.6.9: Conserve boron, sulfur, hydrogen, and oxygen

With coefficients a, b, c, d, the atom-count vector equation yields

$$2a = c, \qquad 3a = d, \qquad 2b = 3c + 2d, \qquad b = 3c.$$

The first, second, and fourth give $c = 2a, d = 3a, b = 6a$. The hydrogen equation checks: $2(6a) = 3(2a) + 2(3a)$. Taking $a = 1$ gives the primitive integer coefficients.

Conclusion. $B_2S_3 + 6H_2O \longrightarrow 2H_3BO_3 + 3H_2S$. The atom totals B, S, H, O are $2, 3, 12, 6$.

Exercise 1.6.10: Use exact fractions in a six-compound reaction

Let the coefficients of the six compounds, in printed order, be a, b, c, d, e, f. The vector equation in atom order Pb, N, Cr, Mn, O has coefficient matrix

$$\begin{bmatrix} 1 & 0 & -3 & 0 & 0 & 0 \\ 6 & 0 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & -2 & 0 & 0 \\ 0 & 2 & 0 & 0 & -1 & 0 \\ 0 & 8 & -4 & -3 & -2 & -1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \\ e \\ f \end{bmatrix} = 0.$$

The first four rows give $a = 3c$, $f = 6a = 18c$, $b = 2d$, and $e = 2b = 4d$. The last row says

$$8b = 4c + 3d + 2e + f \quad \Longrightarrow \quad 16d = 22c + 11d, \quad 5d = 22c.$$

To get positive integers choose $c = 5$, so $d = 22$, $a = 15, b = 44, e = 88, f = 90$. The entries have common divisor 1.

Conclusion. $15PbN_6 + 44CrMn_2O_8 \longrightarrow 5Pb_3O_4 + 22Cr_2O_3 + 88MnO_2 + 90NO$. The oxygen check is $44(8) = 5(4) + 22(3) + 88(2) + 90 = 352$. The other rows already check each remaining element.

Exercise 1.6.11: Balance seven compounds without decimal rounding

Use coefficients a, b, c, d, e, f, g for the seven printed compounds. Conserve Mn, S, As, Cr, H, O, respectively:

$$a = d, \qquad a + c = 3f, \qquad 2b = e,$$
$$10b = f, \qquad 2c = d + 3e + 2g, \qquad 35b + 4c = 4d + 12f + g.$$

These are the coordinates of a single atom-count vector equation. The first four give $d = a$, $e = 2b$, $f = 10b$, $c = 30b - a$. The hydrogen equation becomes $2g = 54b - 3a$. The oxygen equation becomes $g = 35b - 8a$. Equating the two formulas gives

$$54b - 3a = 70b - 16a, \qquad 13a = 16b.$$

Choose $b = 13, a = 16$. Then $c = 374, d = 16, e = 26, f = 130, g = 327$.

$$16\mathrm{MnS} + 13\mathrm{As_2Cr_{10}O_{35}} + 374\mathrm{H_2SO_4} \longrightarrow 16\mathrm{HMnO_4} + 26\mathrm{AsH_3} + 130\mathrm{CrS_3O_{12}} + 327\mathrm{H_2O}.$$

Check sulfur: $16 + 374 = 390 = 3(130)$; hydrogen: $2(374) = 748 = 16 + 3(26) + 2(327)$; oxygen: $35(13) + 4(374) = 1951 = 4(16) + 12(130) + 327$. Mn, As, and Cr match by the first equations.

Conclusion. The displayed positive integer coefficients balance every atom; their common divisor is 1.

Exercise 1.6.12: Use junction equations before imposing nonnegative flow

Read the arrows at the three nodes. Node A receives $x_1 + x_4$ and sends x_2; node B receives x_2 and sends $x_3 + 100$; node C receives $x_3 + 80$ and sends x_4. Thus

$$x_1 + x_4 = x_2, \qquad x_2 = x_3 + 100, \qquad x_4 = x_3 + 80.$$

The last two inserted in the first give $x_1 = 20$. Set $x_3 = t$. The general algebraic flow is

$$(x_1, x_2, x_3, x_4) = (20, 100 + t, t, 80 + t).$$

Nonnegative flows require $t \geq 0$; that condition makes every other component nonnegative too. Hence $x_4 = 80 + t \geq 80$, and $t = 0$ attains the lower bound.

Conclusion. The smallest possible x_4 is 80.

Exercise 1.6.13: A five-node network with two independent free flows

(a) Record all node balances.

Following the arrows in the supplied figure gives

$$A: \quad 30 + x_2 = 80 + x_1, \qquad E: \quad 60 + x_1 = 20 + x_3,$$
$$B: \quad x_3 + x_5 = x_2 + x_4, \qquad C: \quad 100 + x_6 = 40 + x_5,$$
$$D: \quad 40 + x_4 = 90 + x_6.$$

Let $s = x_1$ and $t = x_6$. The first, second, fourth, and fifth balances give

$$(x_1, x_2, x_3, x_4, x_5, x_6) = (s, s + 50, s + 40, t + 50, t + 60, t).$$

The balance at B checks identically: $(s + 40) + (t + 60) = (s + 50) + (t + 50)$.

(b) Apply the arrow-direction restrictions.

Nonnegative x_1 and x_6 require $s, t \geq 0$, and these inequalities make all other flows nonnegative. Their individual minima are therefore

$$\min x_2 = 50, \quad \min x_3 = 40, \quad \min x_4 = 50, \quad \min x_5 = 60.$$

They can all be attained at once by taking $s = t = 0$.

Exercise 1.6.14: A road closure specializes a two-parameter traffic pattern

(a) Write flow in equals flow out at each intersection.

The four balances read

$$80 = x_1 + x_5, \quad 100 + x_1 + x_2 = x_4, \quad x_3 = x_2 + 90, \quad x_4 + x_5 = x_3 + 90.$$

The last is dependent on the first three; total external inflow $80 + 100$ equals total external outflow $90 + 90$. Put $s = x_2$ and $t = x_5$. Then

$$(x_1, x_2, x_3, x_4, x_5) = (80 - t, s, 90 + s, 180 + s - t, t).$$

For the indicated traffic directions require $s \geq 0$ and $0 \leq t \leq 80$; the remaining inequalities then follow.

(b) Close the road carrying x_5.

Setting $t = 0$ gives

$$(x_1, x_2, x_3, x_4, x_5) = (80, s, 90 + s, 180 + s, 0), \qquad s \geq 0.$$

This is the entire feasible family with the road closed, rather than one arbitrarily selected flow pattern.

(c) Minimize x_4.

The formula $x_4 = 180 + s$ with $s \geq 0$ gives the lower bound 180. It is attained when $s = 0$.

Conclusion. With $x_5 = 0$, the minimum flow in branch 4 is 180 cars/minute.

Exercise 1.6.15: Circulating flow around a one-way roundabout

Read the successive balances around A, B, C, D, E, F:

$$x_6 + 60 = x_1, \quad x_1 = x_2 + 70, \quad x_2 + 100 = x_3,$$

$$x_3 = x_4 + 90, \quad x_4 + 80 = x_5, \quad x_5 = x_6 + 80.$$

Choose $t = x_6$. Working around the circle yields

$$(x_1, x_2, x_3, x_4, x_5, x_6) = (t + 60, t - 10, t + 90, t, t + 80, t).$$

The final equation checks because $(t + 80) - t = 80$. To respect every arrow all six flows must be nonnegative. The strongest restriction is $x_2 = t - 10 \geq 0$, namely $t \geq 10$; this also ensures each other component is nonnegative.

Conclusion. All feasible flows are the displayed family with $t \geq 10$, and the smallest possible x_6 is 10.